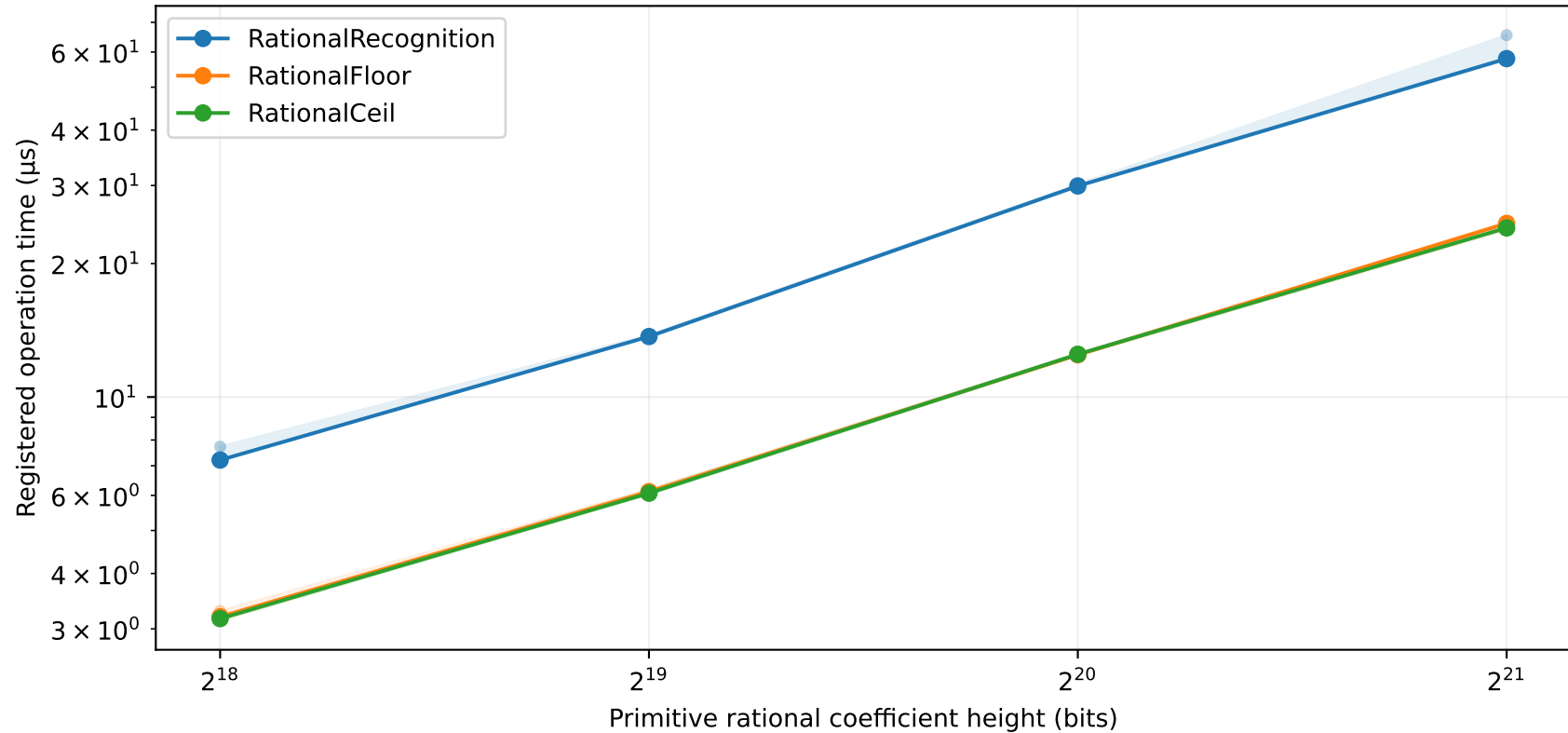


Direct rational leaves after the proved square-root initializer



Four fixed outer trials at each height; every exported point retained. Timers exclude canonical preparation. Canonical preparation remains costly: recognition whole-parent time is about 38 minutes. This source-scoped collection supplies no before/after ratio or general canonical-construction bound. Source 16cff198bb, chungus2, leased CPU 58. All completed arms and failures retained; no load filtering or rerun.